RAJALAKSHMI ENGINEERING COLLEGE

Akash M 250701034-2025-CSE

A2

Started on	Friday, 21 August 2026, 4:14 PM
State	Finished
Completed on	Friday, 21 August 2026, 4:14 PM
Time taken	24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findPair(int arr[], int low, int high, int x, int *a, int *b)
{
    if (low >= high)
        return 0;

    int sum = arr[low] + arr[high];

    if (sum == x)
    {
        *a = arr[low];
        *b = arr[high];
        return 1;
    }

    if (sum < x)
        return findPair(arr, low + 1, high, x, a, b);

    return findPair(arr, low, high - 1, x, a, b);
}

int main()
{
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    int x;
    scanf("%d", &x);

    int a, b;

    if (findPair(arr, 0, n - 1, x, &a, &b))
    {
        printf("%d\n", a);
        printf("%d\n", b);
    }
    else
    {
        printf("No");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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